Computer Design

COA_AFI CLASS TEST

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38/40

1) if ((a<b) and ((c<d) or (a>d))) then
Z = x + y * z
else
Z = Z + 1

Sol:

L1 = a<b goto L2
goto L5

L2 = c<d goto L4
goto L3

L3 = a>d goto L4
goto L5

L4 :

t1 = y * z
t2 = x + t1
Z = t2
goto L6

L5 : Z = Z + 1

L6 :
(exit)

2) (a<b) and (c<d) or (e<f)

Back patching technique

(a<b) and (c<d) goto (e<f)

1. if a<b goto L3

If the first AND expression is false then go (e<f).

...c) into

3.

1. if a<b goto
2. goto 5
3. if c<d goto 7
4. goto 5
5. if e<f goto 7
6. goto 8
7. True
8. False

Back patching means generating jump first

$A = (-A + B) * ((-A + B) * C)$

$t_1 = -A$

$t_2 = t_1 + B$

$t_3 = -A$

$t_4 = t_3 + B$

$t_5 = t_4 * C$

$t_6 = t_5 * t_2$

$A = t_6$

operators	1st Exp	2nd Exp	store.
Uminus	A	-	t_1
+	t_1	B	t_2
Uminus	A	-	t_3
+	t_3	B	t_4
*	t_4	C	t_5
*	t_5	t_2	t_6
=	t_6	-	A

the expression (a+b) * (c+d) + (a+b+c) into

(a>b) or (c<d) or (e<f)

L1:
if a>b goto L TRUE
goto L2

L2:
if c<d goto L TRUE
goto L3

L3:
if e<f goto L TRUE
goto L FALSE

LTRUE:
// Boolean expression is true
goto LEND

LFALSE:
// Boolean expression is false

L END: